Raw AAS / peptide powders (pre-formulation): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for Raw AAS / peptide powders (pre-formulation) (Peptides, SARMs, and Research Chemicals), evaluated on the finished product as sold. No government sets a heavy-metal content limit for this class; every analyte is measured on every lot under mandatory control and escalated against route-appropriate toxicological anchors, never a product-content ceiling. Certification is heavy-metal safety only and does not endorse the efficacy or legality of these products.

1 | How every number on this standard is set

HMTc Method v2.0 — no-regulatory-anchor (control) form. Government-limit lookup returns nothing for this class, so cells resolve to control or a substantiated not_material; ICH Q3D / USP <232> route-appropriate PDEs are toxicological anchors only, never product-content limits.

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	control	measured every lot; no human-food maximum
Cadmium (Cd)	Tier 1	control	measured every lot; no human-food maximum

Total arsenic (tAs)	Tier 2	control	measured every lot; no human-food maximum
Inorganic arsenic (iAs)	Tier 1	control	measured every lot; no human-food maximum
Mercury, total (tHg)	Tier 1	control	measured every lot; no human-food maximum
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	control	measured every lot; no human-food maximum
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	control	measured every lot; no human-food maximum
Hexavalent chromium (Cr-VI)	Tier 2	control	measured every lot; no human-food maximum

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION

Lead (Pb)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls lead by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.
Cadmium (Cd)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls cadmium by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.

Total arsenic (tAs)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls total arsenic by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.
Inorganic arsenic (iAs)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls inorganic arsenic by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.

Mercury, total (tHg)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls total mercury by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.
Methylmercury (MeHg)	n/m	Methylmercury has no formation, source, or reaction pathway in the abiotic synthetic manufacture of peptides, SARMs, anabolic-androgenic steroids, or research chemicals. Methylmercury is produced by microbial biomethylation in aquatic and sediment environments and enters the human food supply through marine bioaccumulation; there is no biomethylation step in chemical synthesis, purification, or reconstitution. Any mercury contamination of these products is inorganic or elemental mercury introduced by reagents, catalysts, or apparatus, and is fully captured by the total-mercury (tHg) control cell. ICH Q3D itself sets the mercury PDE on inorganic mercury as the worst case for pharmaceutical elemental impurities, confirming that inorganic, not organic, mercury is the relevant species for a manufactured product. This is a no-pathway determination (absence of a formation route), corroborated by the total-Hg control that bounds any organomercury fraction.

Nickel (Ni)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls nickel by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.
Tin (Sn)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls tin by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.

Aluminium (Al)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls aluminium by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.
Total chromium (Cr)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls total chromium by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.

Hexavalent chromium (Cr-VI)	control	No jurisdiction sets a lawful product heavy-metal content limit for this unapproved-drug class, so the Method v2.0 min(percentile, ceiling) rule loses its ceiling term and the occurrence (percentile) term is a firewalled staff quantity that must not be published as a product-content limit. HMTc therefore controls hexavalent chromium by mandatory testing rather than a published concentration number: every certified lot is quantified by the reference method, and the result is judged against the route-appropriate toxicological anchor below, not against a fixed ug/kg cap. The absence of any regulatory backstop is precisely why an independent heavy-metal-safety mark carries its highest consumer-protection value for this class.

4 | Cite this standard

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STATUS AND LICENCE

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COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/-research-chem-raw-powders. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-research-chem-raw-powders.